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1. Valid Parentheses

```
bool isValid(string s) {  
    stack<char> st;  
    for (char ch : s) {  
        if (ch == '(' || ch == '{' || ch == '[') {  
            st.push(ch);  
        } else {  
            if (st.empty())  
                return false;  
            if ((ch == ')' && st.top() != '(') ||  
                (ch == '}' && st.top() != '{') ||  
                (ch == ']' && st.top() != '['))  
                return false;  
            st.pop();  
        }  
    }  
    return st.empty();  
}
```

2. Next Greater Element

```
vector<int> nextGreaterElement(vector<int>& nums) {  
    int n = nums.size();  
    vector<int> ans(n, -1);  
    stack<int> st;
```

```

for (int i = n - 1; i >= 0; i--) {
    while (!st.empty() && st.top() <= nums[i]) {
        st.pop();
    }
    if (!st.empty())
        ans[i] = st.top();
    st.push(nums[i]);
}
return ans;
}

```

3. Largest Rectangle in Histogram

```

int largestRectangleArea(vector<int>& heights) {
    stack<int> st;
    int n = heights.size();
    int maxArea = 0;
    for (int i = 0; i <= n; i++) {
        while (!st.empty() && (i == n || heights[st.top()] >= heights[i])) {
            int height = heights[st.top()];
            st.pop();
            int width;
            if (st.empty())
                width = i;
            else
                width = i - st.top() - 1;
            maxArea = max(maxArea, height * width);
        }
        st.push(i);
    }
    return maxArea;
}

```